

All students should watch the videos and go through the readings accompanying this LGA in preparation for this assignment.

Individual self-test questions

All students should answer this set of questions. You will compare your answers with your group during the next lecture period. Note, we may from time to time include trick questions.

1. When studying an algorithm, our two primary concerns should be the _____ and _____ of the algorithm.
2. True or False: algorithms can only be implemented on a computer.
3. Any *computable* function may be computed by an algorithm applying three control structures: _____, _____, and _____. There are other sets of control structures that have equivalent computational power.
4. _____ are truth-valued statements (i.e., logical propositions) about the state of an algorithm at some specific location in the algorithm. Proving these true is often a key step in proving the algorithm as a whole correct.

Group questions

Distribute the following questions across the members of your group. You will share your solutions (and most importantly the *method* of your solutions) during the next lecture period. Divide up the questions so that each question has at least three solutions from different group members.

1. Rewrite the C++ function below, which outputs a *hailstone sequence* for a specified starting integer, in pseudocode. Compare your pseudocode with that of your group. Where different members of the group made different choices in representing operations, debate the merits of your different choices.

```
void sequence(unsigned int n) {  
    cout << n;  
    while (n != 1) {  
        if (n % 2 == 0) n = n / 2;  
        else n = 3 * n + 1;  
        cout << ' ' << n;  
    }  
    cout << endl;  
}
```


3. Consider the algorithm below for averaging the values in an array, expressed in pseudocode (define the mean of an empty array to be zero):

```
function MEAN( $A$ )      //  $A$  is an array, possibly empty, of real numbers
   $mean \leftarrow 0$ 
   $count \leftarrow 0$ 
  for all  $x$  in  $A$  do
     $sum \leftarrow (mean \times count) + x$ 
     $count \leftarrow count + 1$ 
     $mean \leftarrow sum / count$ 
  return  $mean$ 
```

Next, briefly answer the questions below:

- (a) Is this algorithm clear and unambiguous? Would you be able to easily implement this algorithm in a programming language?

- (b) What assumptions does the algorithm make about its inputs, if any? Does it make any other assumptions?

- (c) Can you prove this algorithm is correct using a loop invariant? What would your invariant be?

4. Rewrite the function `MEAN`, whose pseudocode is given in the last question, as C++ code. Compare your code with that of other members of your group. Discuss any differences in assumptions or approaches among your answers.

For Further Practice

Write (in pseudocode) an algorithm for finding the sum of the values in an array. Be sure to note any assumptions. Then, write down a loop invariant and use it to argue the correctness of your algorithm. (We know you can easily write a solution in code; the point here is to practice the formal tools used in writing about algorithms.)